Examining the relationship between individual differences in perspective taking and syntactic priming in monolingual and bilingual speakers

AUTHOR INFORMATION WILL GO HERE AFTER REVIEW

The Interactive Alignment Model claims that during dialogue, people align linguistic representations across various levels, including the syntactic level, to facilitate effective communication (Pickering & Garrod, 2004). As such, it predicts syntactic priming effects (e.g., Branigan et al., 2000), which under this framework, can be conceptualized as a socially-situated language effect. This implies syntactic priming may be related to other social-cognitive abilities, such as perspective-taking (van Zwet & Unsworth, 2024), or the ability to consider another person's thoughts or feelings. This offers a socially-driven explanation for individual differences in the rates of syntactic priming (e.g., Weatherholtz et al., 2014) that purely cognitive mechanisms currently lack (e.g., Chang et al., 2006; Pickering & Branigan, 1998).

In the present study, we seek to understand whether participants' performance on different perspective-taking tasks with varying levels of social involvement predicts the rate at which they show syntactic alignment in a structural priming task. Given previous research that suggests bilinguals perform better on perspective-taking tasks than monolinguals (e.g. Greenberg et al., 2013; Ikizer & Ramírez- Esparza, 2018; Jávor, 2016), we will also investigate whether there are group differences in perspective-taking and syntactic alignment across monolingual and bilingual participants.

Participants (target N = 150) complete three measures of perspective-taking that vary in their degree of social interaction: the Interpersonal Reactivity Index (IRI; Davis 1980), which is a 28-item self-report perspective-taking scale involving no social interaction; the somewhat socially-situated Social Mindfulness paradigm (SoMi; Van Doesum & Van Lange, 2014; see Figure 3), where participants choose from array of items before their partner gets to choose; and the socially-interactive Director's Task (Keysar, 2000; see Figure 2), where a confederate provides instructions to a participant to move objects on an array where not all of the squares are mutually visible. They also complete an interactive syntactic priming task (Branigan et al., 2000; Horton 2014; see Figure 1), in which a participant and confederate take turns describing simple line drawings and finding matching cards from a set. The confederate's card descriptions are scripted to use either a double object or prepositional object structure, and priming is measured by how often the participant's description match the structure heard on the previous trial. Demographic questions, language proficiency and history questionnaires are also adminitered.

We predict perspective-taking scores will be positively associated with priming, and these associations will be stronger for more socially-situated perspective-taking tasks. We also predict higher perspective-taking scores in bilingual participants (e.g., van Zwet & Unsworth, 2024). A multi-level logistic regression will be run in R (using the lme4 package) with the outcome variable of structural match on each priming task trial. Fixed effects will include perspective-taking scores on the IRI, DT, and SoMi; participant and item will be included as crossed random effects. We will also use independent-samples t-tests to determine whether monolingual and bilingual speakers differ in performance on any of our three perspective-taking or syntactic priming measures. Thus far, we have collected data from two participants, but data collection is on-going.

Figure 1

Example DO and PO Prime from the Syntactic Priming Task

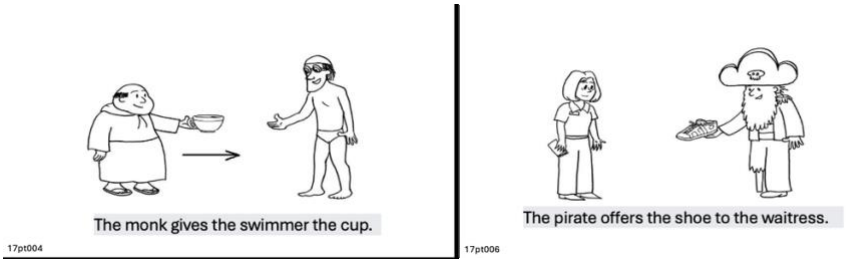


Figure 2

Example of a Target Trial and a Filler Trial from the Director's Task

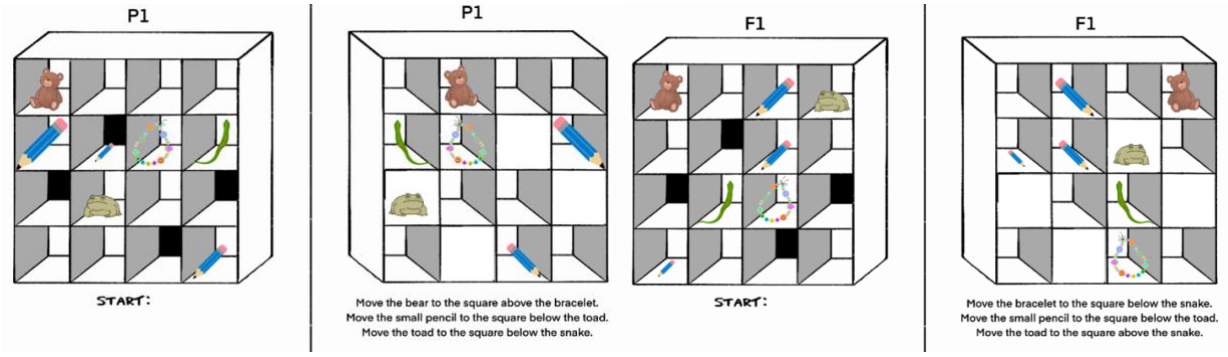
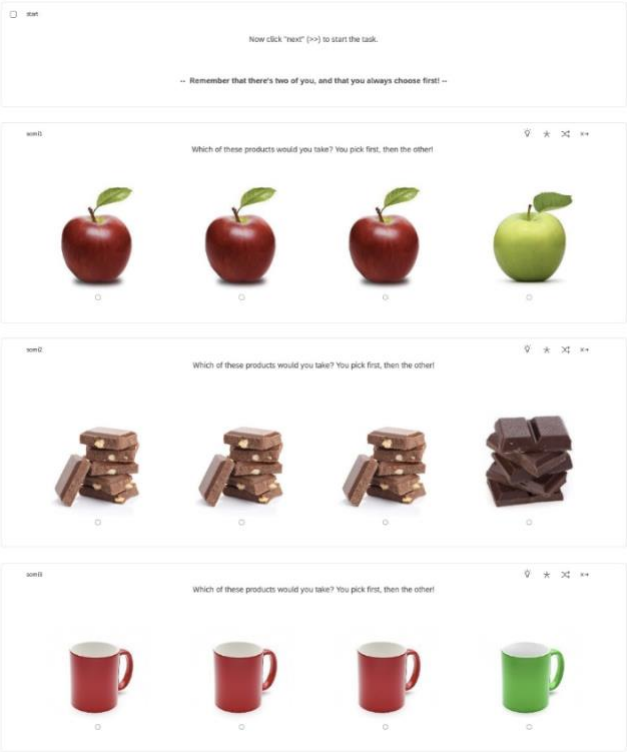


Figure 3

Example of the Social Mindfulness Task Stimuli